

HALF MOON BAY BUOY 46012, USA

WMO: 992480

Lat: 37.356N

Lon: 122.881W

Elev: 0

StdP: 101.33

Time Zone: -8.00 (W08)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	7.9	8.6	N/A	N/A	N/A	N/A	N/A	N/A	15.8	10.2	14.9	10.3	6.4	330	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	1.7	18.0	N/A	17.4	N/A	16.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.6	13.3	11.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	11.5	11.3	11.1	11.1	11.7	12.4	13.4	14.3	14.4	13.9	12.8	11.8
	DBStd	1.81	1.37	1.44	1.30	1.11	1.20	1.08	1.32	1.50	1.54	1.42	1.29	1.46
	HDD10.0	16	3	3	4	3	0	0	0	0	0	0	0	3
	HDD18.3	2136	212	196	224	217	207	179	153	126	118	137	165	202
	CDD10.0	923	49	41	38	37	52	71	105	133	133	122	85	58
	CDD18.3	1	0	0	0	0	0	0	0	0	1	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.7	5.8	6.5	6.3	6.6	6.3	6.1	4.8	4.7	4.4	5.0	5.5	6.0
Precipitation	PrecAvg	638	126	119	84	44	21	7	1	2	3	33	69	131
	PrecMax	1071	335	396	248	147	87	40	2	20	13	109	204	332
	PrecMin	326	8	1	5	5	0	0	0	0	0	1	4	13
	PrecStd	175	90	81	70	33	22	9	0	4	3	29	48	89
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	15.6	15.2	14.6	16.4	15.8	17.7	18.4	22.1	18.4	16.9	16.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	14.5	14.9	14.1	13.6	14.9	15.1	16.9	17.9	18.3	17.5	16.0	15.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	13.9	14.0	13.5	13.0	14.1	14.6	16.2	17.4	17.2	16.9	15.3	14.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	13.3	13.3	12.9	12.7	13.4	14.1	15.4	16.7	16.4	16.2	14.7	13.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	1.8	1.7	1.7	1.5	1.4	1.4	1.5	1.7	1.7	1.8	1.8	1.8
		MCDBR	2.2	2.4	2.1	1.9	2.1	1.9	1.7	1.9	2.1	2.6	2.5	2.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.322	0.333	0.341	0.363	0.368	0.359	0.371	0.342	0.347	0.338	0.321	0.325	
	taud	2.584	2.550	2.529	2.433	2.447	2.475	2.452	2.540	2.498	2.568	2.593	2.582	
	Ebn at Noon	871	904	926	917	913	918	903	927	906	886	866	842	
	Edh at Noon	75	87	97	112	113	110	111	100	98	84	73	71	
All-Sky Solar Radiation	RadAvg	2.43	3.34	4.68	6.03	6.64	7.09	6.10	5.44	4.99	3.87	2.82	2.12	
	RadStd	0.29	0.38	0.37	0.44	0.54	0.41	0.38	0.43	0.35	0.26	0.19	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (38 neighbors)	N/A	N/A	-1.21	N/A	N/A	N/A	N/A	-110	+136	N/A

Nomenclature: See separate page